```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
import zipfile
import os

# Show the files uploaded to Colab
print(os.listdir('/content'))
```

```
['.config', 'archive (2).zip', 'sample_data']
```

```
import zipfile
import os

zip_path = '/content/archive (2).zip'

with zipfile.ZipFile(zip_path, 'r') as zip_ref:
    zip_ref.extractall('/content/shopping_data')

print(os.listdir('/content/shopping_data'))
```

```
['shopping_trends.csv', 'shopping_trends_updated.csv']
```

```
import pandas as pd

df = pd.read_csv('/content/shopping_data/shopping_trends_updated.csv')

df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Di A
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	

Next steps: Generate code with df New interactive sheet

```
print("Dataset Shape:", df.shape)
```

```
Dataset Shape: (3900, 18)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3900 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
```

```
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
print(df.columns.tolist())
```

```
, 'Subscription Status', 'Shipping Type', 'Discount Applied', 'Promo Code Used', 'Previous Purchases', 'Payment Method', 'Freque
```

```
missing_values = df.isnull().sum()

print("Missing Values in Each Column:")
print(missing_values)

print("\nTotal Missing Values:", missing_values.sum())
```

```
Missing Values in Each Column:
Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
Purchase Amount (USD)     0
Location                  0
Size                      0
Color                     0
Season                    0
Review Rating             0
Subscription Status       0
Shipping Type             0
Discount Applied          0
Promo Code Used           0
Previous Purchases        0
Payment Method            0
Frequency of Purchases    0
dtype: int64

Total Missing Values: 0
```

```
duplicate_count = df.duplicated().sum()

print("Number of Duplicate Rows:", duplicate_count)
```

```
Number of Duplicate Rows: 0
```

```
df.describe()
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3900.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.749949	25.351538
std	1125.977353	15.207589	23.685392	0.716223	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.700000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

```
total_purchase = df['Purchase Amount (USD)'].sum()
average_purchase = df['Purchase Amount (USD)'].mean()
median_purchase = df['Purchase Amount (USD)'].median()

print(f"Total Purchase Amount: ${total_purchase:,.2f}")
print(f"Average Purchase Amount: ${average_purchase:.2f}")
print(f"Median Purchase Amount: ${median_purchase:.2f}")
```

```
Total Purchase Amount: $233,081.00
Average Purchase Amount: $59.76
Median Purchase Amount: $60.00
```

```
category_sales = (
    df.groupby('Category')['Purchase Amount (USD)']
    .agg(['sum', 'mean', 'count'])
    .sort_values('sum', ascending=False)
)

category_sales
```

Category	sum	mean	count
Clothing	104264	60.025331	1737
Accessories	74200	59.838710	1240
Footwear	36093	60.255426	599
Outerwear	18524	57.172840	324

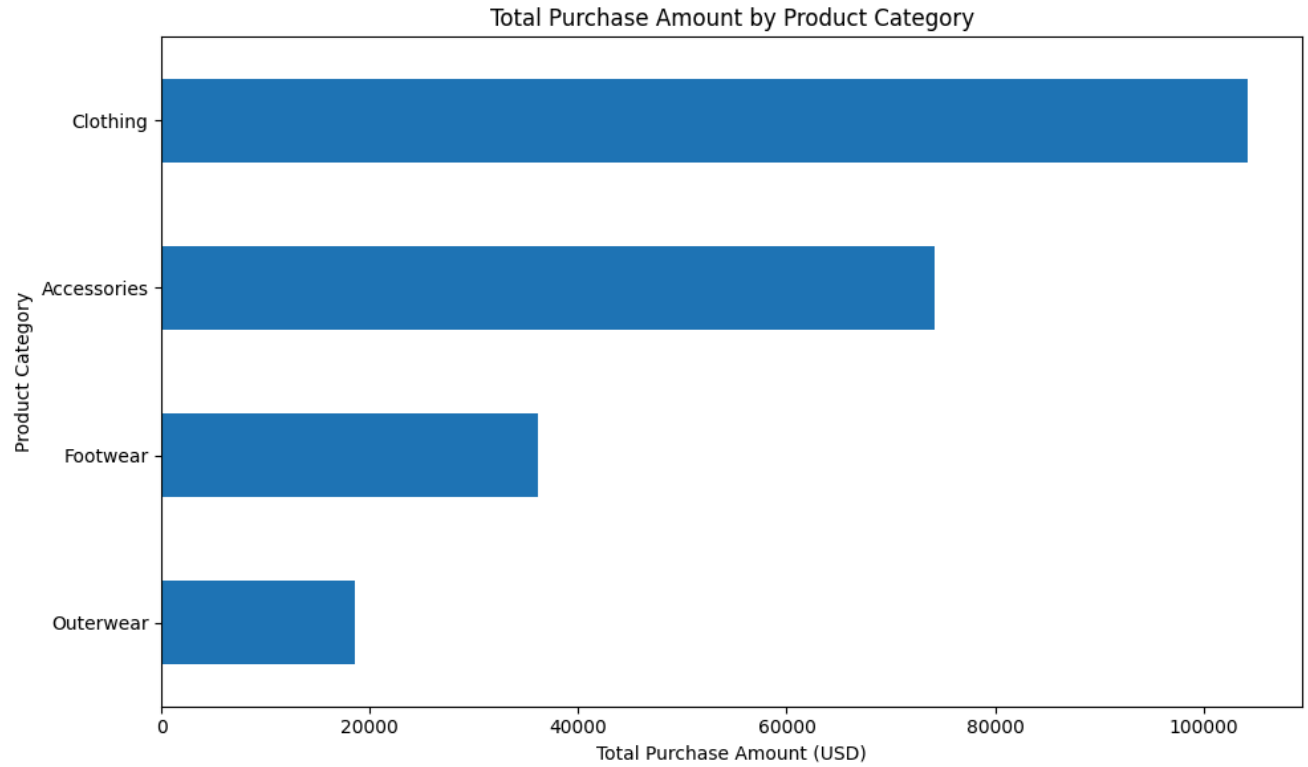
Next steps: Generate code with `category_sales` New interactive sheet

```
plt.figure(figsize=(10, 6))

category_sales['sum'].sort_values().plot(kind='barh')

plt.title('Total Purchase Amount by Product Category')
plt.xlabel('Total Purchase Amount (USD)')
plt.ylabel('Product Category')

plt.tight_layout()
plt.show()
```




```
top_products = df['Item Purchased'].value_counts().head(10)

print("Top 10 Most Frequently Purchased Products:")
print(top_products)
```

```
Top 10 Most Frequently Purchased Products:
Item Purchased
Blouse        171
Pants         171
Jewelry       171
```

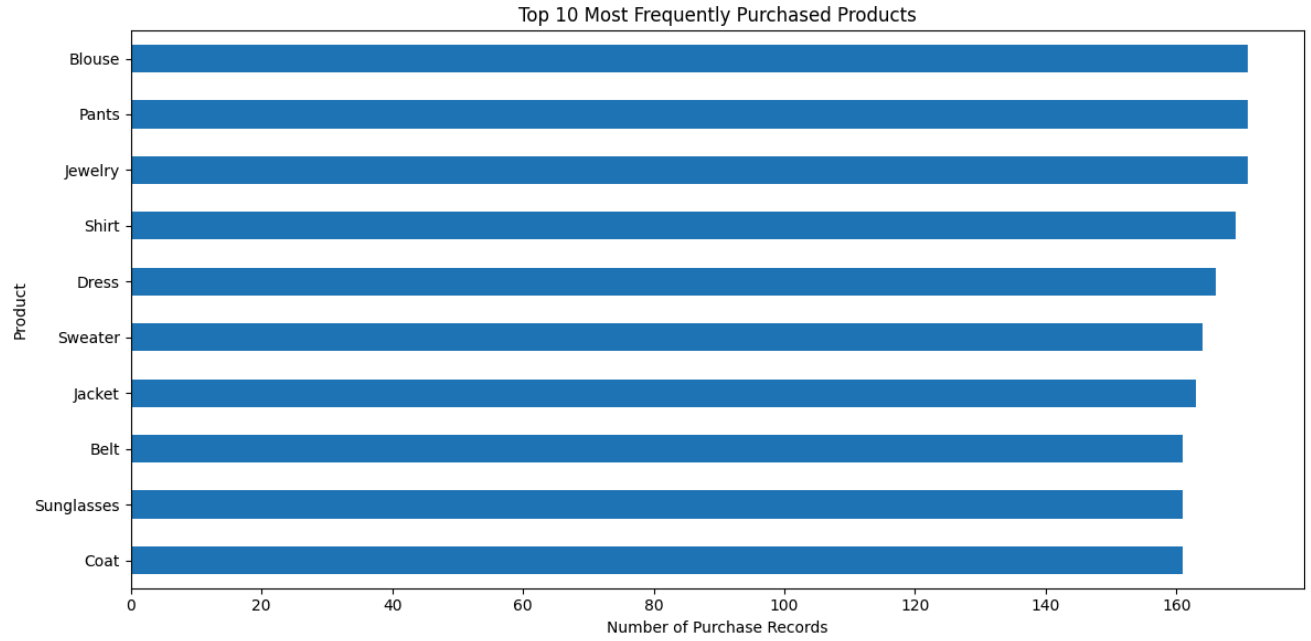
```
Shirt         169
Dress         166
Sweater       164
Jacket        163
Coat          161
Sunglasses    161
Belt          161
Name: count, dtype: int64
```

```
plt.figure(figsize=(12, 6))

top_products.sort_values().plot(kind='barh')

plt.title('Top 10 Most Frequently Purchased Products')
plt.xlabel('Number of Purchase Records')
plt.ylabel('Product')

plt.tight_layout()
plt.show()
```




```
payment_counts = df['Payment Method'].value_counts()

print("Purchase Records by Payment Method:")
print(payment_counts)
```

```
Purchase Records by Payment Method:
Payment Method
PayPal           677
Credit Card      671
Cash             670
Debit Card       636
Venmo            634
Bank Transfer    612
Name: count, dtype: int64
```

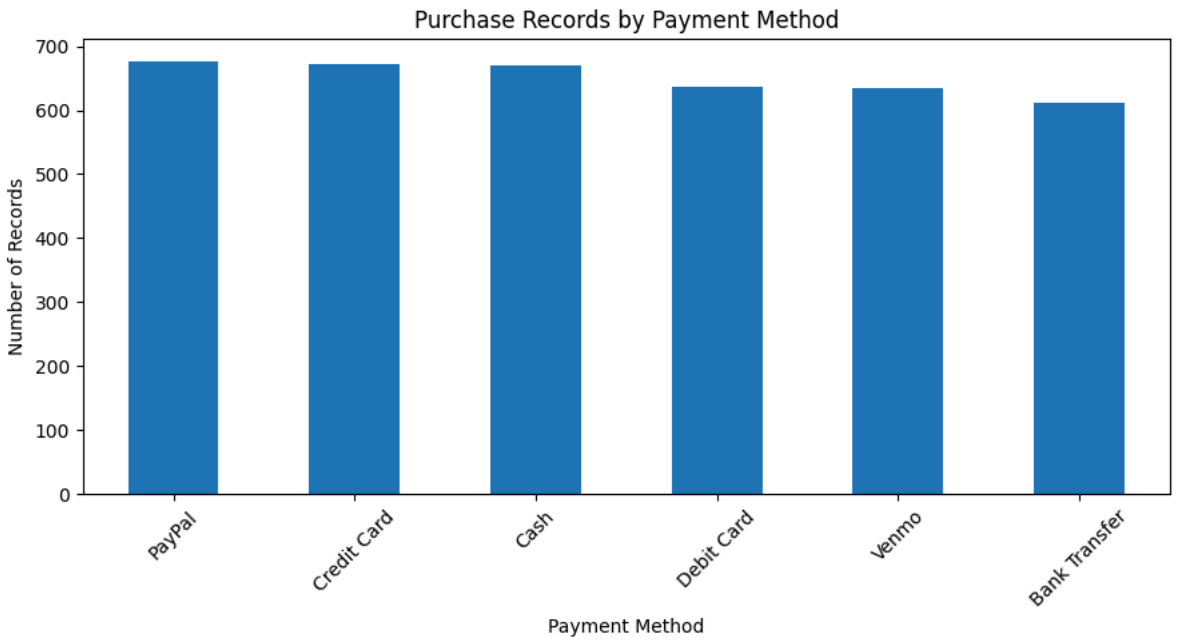
```
plt.figure(figsize=(9, 5))

payment_counts.plot(kind='bar')

plt.title('Purchase Records by Payment Method')
plt.xlabel('Payment Method')
plt.ylabel('Number of Records')

plt.xticks(rotation=45)
plt.tight_layout()
```

```
plt.show()
```




```
gender_analysis = df.groupby('Gender').agg(
    Number_of_Records=('Customer ID', 'count'),
    Average_Purchase=('Purchase Amount (USD)', 'mean'),
    Total_Purchase=('Purchase Amount (USD)', 'sum')
)

gender_analysis
```

Gender	Number_of_Records	Average_Purchase	Total_Purchase
Female	1248	60.249199	75191
Male	2652	59.536199	157890

Next steps: Generate code with gender_analysis | New interactive sheet

```
df['Age Group'] = pd.cut(
    df['Age'],
    bins=[0, 20, 30, 40, 50, 100],
    labels=['Under 20', '20-30', '31-40', '41-50', 'Above 50'],
    include_lowest=True
)

age_analysis = df.groupby(
    'Age Group',
    observed=False
).agg(
    Number_of_Records=('Customer ID', 'count'),
    Average_Purchase=('Purchase Amount (USD)', 'mean'),
    Total_Purchase=('Purchase Amount (USD)', 'sum')
)

age_analysis
```

Age Group	Number_of_Records	Average_Purchase	Total_Purchase
Under 20	212	58.981132	12504
20-30	737	60.753053	44775
31-40	725	59.576552	43193
41-50	750	58.838667	44129
Above 50	1476	59.945799	88480

Next steps: Generate code with age_analysis New interactive sheet

```
season_analysis = df.groupby('Season').agg(
    Number_of_Records=('Customer ID', 'count'),
    Average_Purchase=('Purchase Amount (USD)', 'mean'),
    Total_Purchase=('Purchase Amount (USD)', 'sum')
).sort_values('Total_Purchase', ascending=False)

season_analysis
```

Season	Number_of_Records	Average_Purchase	Total_Purchase
Fall	975	61.556923	60018
Spring	999	58.737738	58679
Winter	971	60.357364	58607
Summer	955	58.405236	55777

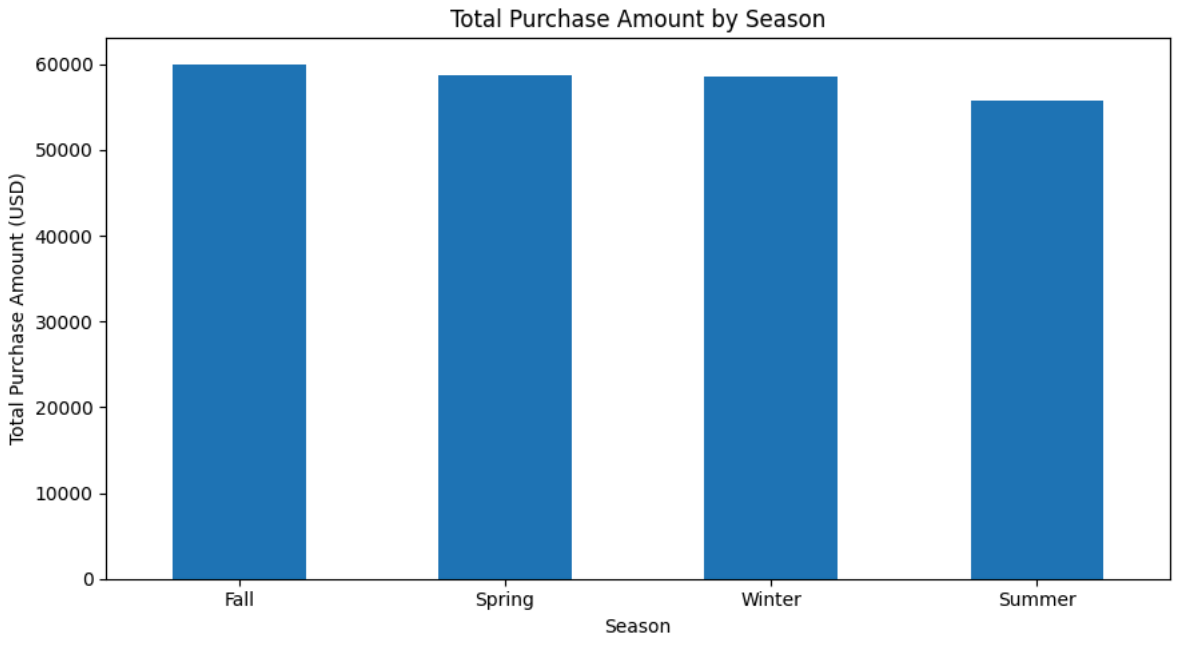
Next steps: Generate code with season_analysis New interactive sheet

```
plt.figure(figsize=(9, 5))

season_analysis['Total_Purchase'].plot(kind='bar')

plt.title('Total Purchase Amount by Season')
plt.xlabel('Season')
plt.ylabel('Total Purchase Amount (USD)')

plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



```
print("Average Review Rating:", df['Review Rating'].mean())

print("\nReview Rating Statistics:")
print(df['Review Rating'].describe())
```

```
Average Review Rating: 3.7499487179487176

Review Rating Statistics:
count    3900.000000
mean        3.749949
std         0.716223
min         2.500000
25%         3.100000
50%         3.700000
75%         4.400000
max         5.000000
Name: Review Rating, dtype: float64
```

```
discount_analysis = df.groupby('Discount Applied').agg(
    Number_of_Records=('Customer ID', 'count'),
    Average_Purchase=('Purchase Amount (USD)', 'mean'),
    Total_Purchase=('Purchase Amount (USD)', 'sum')
)

discount_analysis
```

Discount Applied	Number_of_Records	Average_Purchase	Total_Purchase
No	2223	60.130454	133670
Yes	1677	59.279070	99411

Next steps: Generate code with `discount_analysis` New interactive sheet

```
subscription_analysis = df.groupby('Subscription Status').agg(
    Number_of_Records=('Customer ID', 'count'),
    Average_Purchase=('Purchase Amount (USD)', 'mean'),
    Total_Purchase=('Purchase Amount (USD)', 'sum')
)

subscription_analysis
```

Subscription Status	Number_of_Records	Average_Purchase	Total_Purchase
No	2847	59.865121	170436
Yes	1053	59.491928	62645

Next steps: Generate code with `subscription_analysis` New interactive sheet

```
# ========================================
# CUSTOMER SHOPPING ANALYSIS DASHBOARD
# ========================================

import pandas as pd
import matplotlib.pyplot as plt
from IPython.display import display, Markdown

# 1. Calculate Key Performance Indicators (KPIs)

total_customers = df['Customer ID'].nunique()
total_records = len(df)
total_purchase = df['Purchase Amount (USD)'].sum()
average_purchase = df['Purchase Amount (USD)'].mean()
average_rating = df['Review Rating'].mean()

# 2. Display Dashboard Title

display(Markdown("# Customer Shopping Analysis Dashboard"))
display(Markdown("### Key Performance Indicators (KPIs)"))

print(f"Total Customers: {total_customers:,}")
```

```
print(f"Total Shopping Records: {total_records:,}")
print(f"Total Purchase Amount: ${total_purchase:,.2f}")
print(f"Average Purchase Amount: ${average_purchase:.2f}")
print(f"Average Review Rating: {average_rating:.2f} / 5")

# 3. Purchase Amount by Category

category_sales = df.groupby('Category')['Purchase Amount (USD)'].sum()
category_sales = category_sales.sort_values(ascending=False)

plt.figure(figsize=(9, 5))
category_sales.plot(kind='bar')
plt.title('Total Purchase Amount by Category')
plt.xlabel('Category')
plt.ylabel('Purchase Amount (USD)')
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()

# 4. Customer Distribution by Gender

gender_counts = df['Gender'].value_counts()

plt.figure(figsize=(7, 5))
gender_counts.plot(kind='bar')
plt.title('Customer Records by Gender')
plt.xlabel('Gender')
plt.ylabel('Number of Records')
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

# 5. Shopping Records by Season

season_counts = df['Season'].value_counts()

plt.figure(figsize=(7, 5))
season_counts.plot(kind='bar')
plt.title('Shopping Records by Season')
plt.xlabel('Season')
plt.ylabel('Number of Records')
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

# 6. Average Purchase by Age Group

if 'Age Group' not in df.columns:
    df['Age Group'] = pd.cut(
        df['Age'],
        bins=[0, 20, 30, 40, 50, 100],
        labels=['Under 20', '20-30', '31-40', '41-50', 'Above 50'],
        include_lowest=True
    )

age_purchase = df.groupby(
    'Age Group', observed=False
)['Purchase Amount (USD)'].mean()

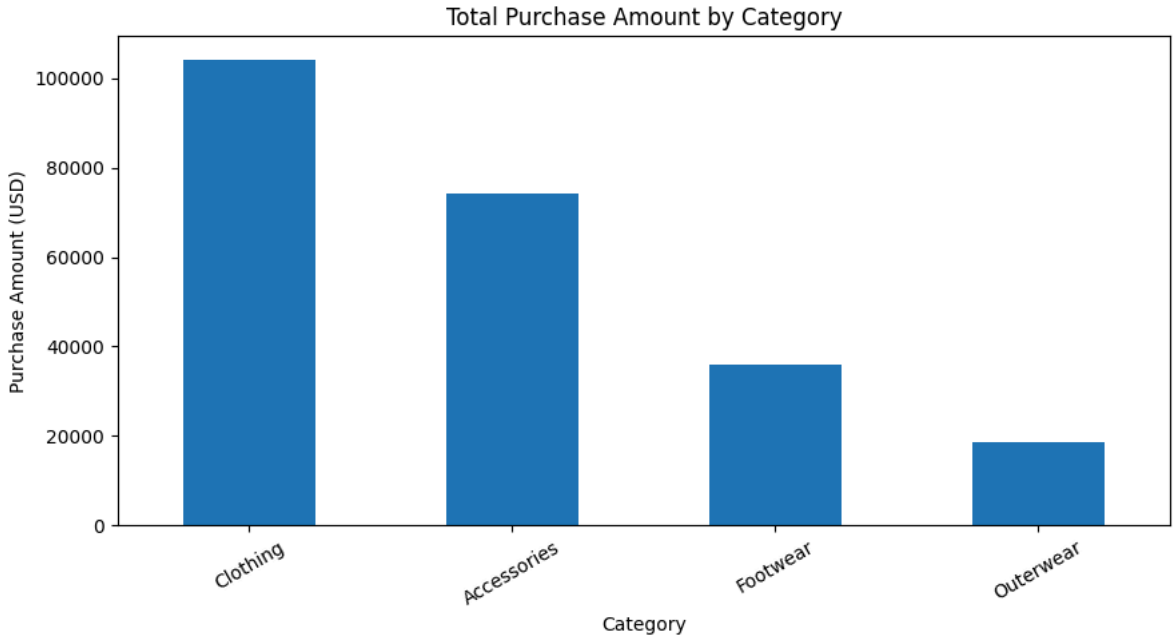
plt.figure(figsize=(9, 5))
age_purchase.plot(kind='bar')
plt.title('Average Purchase Amount by Age Group')
plt.xlabel('Age Group')
plt.ylabel('Average Purchase Amount (USD)')
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

print("\nDashboard completed successfully!")
```

Customer Shopping Analysis Dashboard

Key Performance Indicators (KPIs)

```
Total Customers: 3,900
Total Shopping Records: 3,900
Total Purchase Amount: $233,081.00
Average Purchase Amount: $59.76
Average Review Rating: 3.75 / 5
```






```
# ========================================
# BUSINESS INSIGHTS
# ========================================

print("=" * 50)
print("CUSTOMER SHOPPING BUSINESS INSIGHTS")
print("=" * 50)

# Insight 1: Best-selling category by total purchase amount
category_sales = df.groupby('Category')['Purchase Amount (USD)'].sum()
best_category = category_sales.idxmax()

print("\n1. Top Category by Purchase Amount:")
print(f"{best_category}: ${category_sales.max():,.2f}")

# Insight 2: Most frequently purchased item
popular_item = df['Item Purchased'].value_counts().idxmax()
popular_item_count = df['Item Purchased'].value_counts().max()

print("\n2. Most Frequently Recorded Item:")
print(f"{popular_item}: {popular_item_count} records")

# Insight 3: Most common payment method
payment_counts = df['Payment Method'].value_counts()
popular_payment = payment_counts.idxmax()

print("\n3. Most Common Payment Method:")
print(f"{popular_payment}: {payment_counts.max()} records")

# Insight 4: Highest-spending season by total purchase amount
season_sales = df.groupby('Season')['Purchase Amount (USD)'].sum()
best_season = season_sales.idxmax()

print("\n4. Season with the Highest Total Purchase Amount:")
print(f"{best_season}: ${season_sales.max():,.2f}")

# Insight 5: Subscription status distribution
subscription_counts = df['Subscription Status'].value_counts()
```

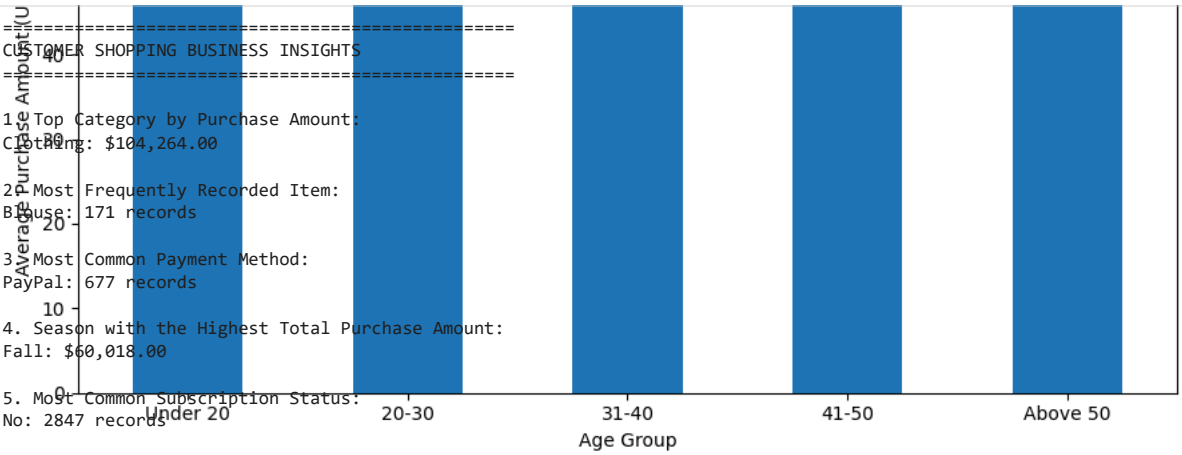
```
popular_subscription_status = subscription_counts.idxmax()

print("\n5. Most Common Subscription Status:")
print(
    f"{popular_subscription_status}: "
    f"{subscription_counts.max()} records"
)

# Insight 6: Overall customer review rating
average_rating = df['Review Rating'].mean()

print("\n6. Average Review Rating:")
print(f"{average_rating:.2f} out of 5")

print("\n" + "=" * 50)
print("BUSINESS INSIGHTS GENERATED SUCCESSFULLY!")
print("=" * 50)
```




```
==================================================
CUSTOMER SHOPPING BUSINESS INSIGHTS
==================================================

1. Top Category by Purchase Amount:
Clothing: $104,264.00

2. Most Frequently Recorded Item:
Blouse: 171 records

3. Most Common Payment Method:
PayPal: 677 records

4. Season with the Highest Total Purchase Amount:
Fall: $60,018.00

5. Most Common Subscription Status:
No: 2847 records

6. Average Review Rating:
3.75 out of 5

Dashboard completed successfully!

==================================================
BUSINESS INSIGHTS GENERATED SUCCESSFULLY!
==================================================
```

Close

```
# Customer Shopping Analysis Using Python

## Project Overview

This project analyzes customer shopping trends using Python to ide
purchasing patterns, popular products, payment preferences, season
and customer subscription behavior.

## Tools and Libraries

* Python
* Pandas
* NumPy
* Matplotlib
* Seaborn
* Google Colab

## Dataset

The project uses the Customer Shopping Trends Dataset, containing
and 18 original columns related to customer demographics, purchase
spending amounts, payment methods, seasons, and customer reviews.

## Key Findings

* Clothing recorded the highest total purchase amount at $104,264.
* Blouse was the most frequently recorded item, with 171 records.
* PayPal was the most frequently recorded payment method, appearin
records.
* Fall had the highest total purchase amount among the seasons, at
* The most common subscription status was "No," appearing in 2,847
* The average customer review rating was 3.75 out of 5.
```

Customer Shopping Analysis Using Python

Project Overview

This project analyzes customer shopping trends using Python to identify purchasing patterns, popular products, payment preferences, seasonal trends, and customer subscription behavior.

Tools and Libraries

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

Dataset

The project uses the Customer Shopping Trends Dataset, containing 3,900 records and 18 original columns related to customer demographics, purchased products, spending amounts, payment methods, seasons, and customer reviews.

```
## Analysis Performed

* Data inspection and quality checks
* Summary statistics
* Purchase analysis by product category
* Product popularity analysis
* Payment method analysis
* Customer analysis by gender and age group
* Seasonal shopping analysis
* Customer review analysis
* Discount and subscription analysis
* Dashboard visualization
* Business insights

## Conclusion

The analysis highlights differences in purchasing activity across
categories, seasons, and payment methods. These findings can help
investigate product demand, improve marketing campaigns, and devel
to encourage customer subscriptions.

## Author

Habiba Mohmed
```

Key Findings

- Clothing recorded the highest total purchase amount at $104,264$.
  $*Blousewasthemostfrequentlyrecordeditem,$
  $with171records.$
  $*PayPalwasthemostfrequentlyrecordedpaymentmeth$
  $appearingin677records.$
  $*Fallhadthehighesttotalpurchaseamountamongthesea$
  at
  60,018.
- The most common subscription status was "No," appearing in 2,847 records.
- The average customer review rating was 3.75 out of 5.

Analysis Performed

- Data inspection and quality checks
- Summary statistics
- Purchase analysis by product category
- Product popularity analysis
- Payment method analysis
- Customer analysis by gender and age group
- Seasonal shopping analysis
- Customer review analysis
- Discount and subscription analysis
- Dashboard visualization
- Business insights

Conclusion

The analysis highlights differences in purchasing activity across product categories, seasons, and payment methods. These findings can help a retailer investigate product demand, improve marketing campaigns, and develop strategies to encourage customer subscriptions.

Author

Habiba Mohmed